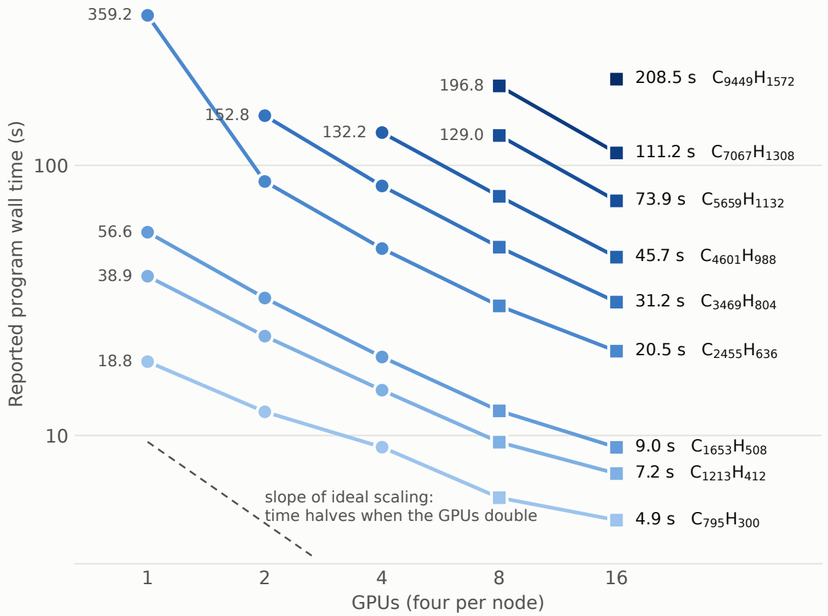
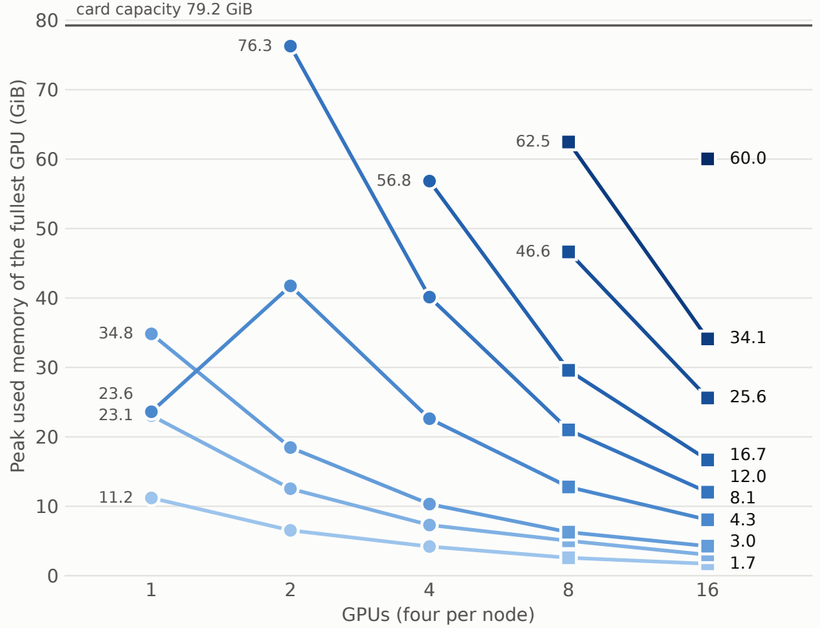
Complete SCF calculations on 1 to 16 GPUs: time, GPU memory and host memory

$C_{795}H_{300}$ (3,480 e) — $C_{1213}H_{412}$ (5,264 e) — $C_{1653}H_{508}$ (7,120 e)
$C_{2455}H_{636}$ (10,456 e) — $C_{3469}H_{804}$ (14,680 e)
$C_{4601}H_{988}$ (19,392 e) — $C_{5659}H_{1132}$ (23,768 e)
$C_{7067}H_{1308}$ (29,576 e) — $C_{9449}H_{1572}$ (39,368 e)
one array per sector
basis shared by the GPUs of a sector

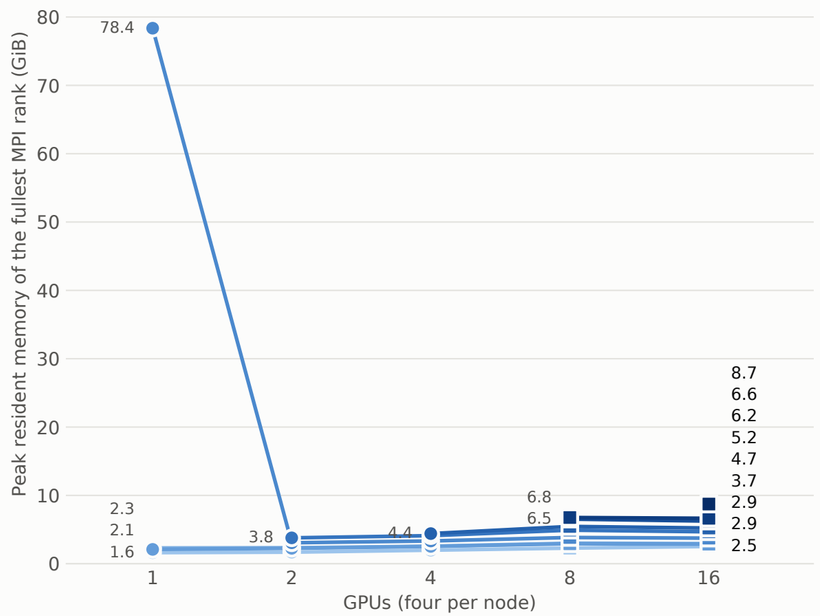
Complete SCF time



GPU memory



Host memory (one rank per node)



m-up (C2455H636 on 1 GPU, C3469H804 on 2 GPUs: the second of two runs on the node); against series F13 (FF_MaxIter 4): program time -2 to -18 % in 31 of 32 cells, k09 on 1 GPU +3.1 %; totals within 1.1e-4 Ry. GPU memory: nvidia-smi samples every 200 ms, fullest GPU, including contexts and allocator caches. Host memory: resident hi
Series F14 (5 A of vacuum): code defaults, double precision, first calculations without symmetry cache. Source 73d5947.